

Implicit Content-Domain: The Anti-Pattern That Arises When Entities Operate Without Authored Content-Domain Specifications per B1.18

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026).

Abstract

B1.18 establishes content-domain as the authored operational territory of each entity in a CKS architecture—the specification, authored through substrate content, of what inputs an entity handles, what outputs it produces, and where its boundaries lie. The Implicit Content-Domain anti-pattern arises when this specification is absent, too vague to govern actual boundaries, or not enforced through boundary rules in entity DNA. The pattern presents in three recognizable forms: absent content-domain, where no specification exists at all; vague content-domain, where specification exists but is too imprecise to derive enforcement from; and unenforced content-domain, where specification exists and is specific but boundary enforcement rules are missing from DNA. All three forms share the same structural failure: operational territory is not in substrate. The consequences follow from that absence—scope drifts without governance, composition compatibility cannot be assessed, governance cannot inspect what it cannot see, and action-feedback cannot distinguish in-domain behavioral patterns from out-of-domain processing. This note formalizes the anti-pattern, traces its three recognizable forms, identifies the emergence conditions that produce each, catalogs the operational consequences, specifies the detection checks, and describes the remediation path through directed selection per B1.14.

1. The Commitment This Anti-Pattern Violates

B1.18 establishes content-domain as a required authored specification for every entity in the architecture. For cells, content-domain is expressed through input domain schemas per B2.25—authored specifications of what inputs the cell accepts, what constitutes an

out-of-domain request, and how domain membership is determined at execution time. For aspects, content-domain is expressed through purpose statements per B2.15—authored specifications of the aspect’s operational territory across its cells. For Selves, content-domain is expressed through integration scope per B2.21—authored specifications of what the Self handles at the whole-organization level and where its boundaries are.

The commitment is not merely that entities have a purpose or a function that developers understand. B1.18 requires that content-domain be *authored through substrate content*—present in the entity’s DNA per B2.25, inspectable by governance, versioned as substrate content changes, and derivable into boundary enforcement rules per B2.91 that can be tested at execution time. Content-domain is a governance artifact, not a developer assumption.

B2.89 through B2.93 decompose B1.18. B2.90 specifies the requirements for content-domain specification at each level. B2.91 specifies how authored content-domain translates into boundary enforcement rules in entity DNA. B2.92 establishes that content-domain specifications are the basis for composition compatibility assessment—two entities can only be assessed for composition fit if both carry authored content-domain specifications that can be compared. B2.93 establishes content-domain verification as the operational test that confirms completeness, boundary coherence, and composition compatibility across the entities in scope.

The Implicit Content-Domain anti-pattern is the failure to satisfy this commitment in any of three forms.

2. Recognizable Form

The anti-pattern presents in three structurally distinct forms. Each form represents a different distance from compliance, but all three share the fundamental failure: operational territory is not authoritatively present in substrate in a form that governance can use.

Form 1 — Absent Content-Domain

The entity has no authored content-domain specification at all. For a cell, this means no input domain schemas per B2.25 exist in the DNA layer—there is no authored record of what the cell handles. For an aspect, no purpose statement per B2.15 has been authored—the aspect’s operational territory is undefined in substrate. For a Self, no integration scope per B2.21 has been specified—the Self’s overall domain exists only as an implicit assumption held by the humans who built it.

The recognition signals are direct. B2.90 content-domain specification requirements are unmet—the required elements are simply absent. B2.93 content-domain verification fails on specification completeness before it can proceed to boundary coherence or composition compatibility. The entity’s birth specification per B2.40 was incomplete: content-domain

elements were not authored when the entity was originated, and no subsequent directed selection per B1.14 has remediated the gap. Composition compatibility per B2.92 cannot be assessed—there is no authored domain to compare with peers.

This form is the starkest instance of the anti-pattern. The entity operates, but its operational territory is known only to whoever built it. That knowledge is not in substrate, is not subject to governance rights per A1.01, and cannot be inspected, modified, or overridden as architectural commitments require. If the developer who understood the implicit scope is unavailable, the scope is unknown to the system.

Form 2 — Vague Content-Domain

The entity has an authored content-domain specification, but the specification is written too broadly or imprecisely to govern actual boundaries. A cell specification that reads “handles all customer-related queries” exists as authored content, but it cannot answer the question governance needs it to answer: for a given input X, is that input in this entity’s domain or not? An aspect specification that reads “covers all product operations” cannot be used to derive enforcement boundaries. A Self specification that reads “supports the enterprise’s coordination needs” carries no operational content that can be turned into boundary rules.

The recognition signals are diagnostic rather than simply absent. The content-domain specification per B2.90 exists—the check for presence passes—but the specificity check fails. B2.91 boundary enforcement rules are absent from entity DNA, not because the specification was omitted but because the specification is too vague to derive boundaries from. Boundary conflicts per B2.91 are never registered in substrate because boundaries are never clear enough to detect violations. A2.40 out-of-domain processing violations never fire because the domain specification does not define what “out of domain” means with enough precision to evaluate against.

This form is architecturally more dangerous than Form 1 in one respect: it passes superficial review. The specification artifact exists; its presence is visible in substrate; the entity appears to have a governed content-domain. The failure is in the specification’s operational uselessness—it cannot do the work the architecture requires of it, but nothing in a surface inspection reveals that failure without applying the specificity check from B2.90.

Form 3 — Unenforced Content-Domain

The entity has an authored content-domain specification that is specific enough to govern boundaries—the specification, if used, could answer whether a given input is in or out of domain. But boundary enforcement rules per B2.91 are absent from the entity’s DNA. The specification sits in substrate as authored content, but no operational mechanism checks inputs against it at execution time.

The recognition signals combine specification presence with enforcement absence. B2.91 boundary enforcement rules are absent from DNA per B2.25—they were never authored, or they were authored and subsequently lost during a DNA evolution event. No out-of-domain handling rules per B2.90 exist to route, reject, or flag inputs that fall outside the authored domain. Entities receive any inputs—including inputs clearly outside their authored domain—without boundary violation recording per A2.40. B2.91 boundary violation detection never fires not because boundaries are unclear but because no detection mechanism is present.

This form represents a governance architecture that was designed but not completed. The intent to specify content-domain is visible in the authored specification. The implementation of that intent—the enforcement rules that make the specification operationally active—was not authored into the entity’s DNA. The specification governs nothing because nothing executes against it.

3. Emergence Conditions

Three conditions reliably produce the Implicit Content-Domain anti-pattern, each associated most strongly with one of the three forms.

Scope as implementation detail. Architects treat content-domain as an internal concern of the entity’s implementation, not as a governance specification requiring authored substrate content. The reasoning takes the form: “the implementation handles the right inputs; specifying what those inputs are in governance terms adds overhead without adding capability.” This condition produces Form 1—absent content-domain—because scope is never authored into substrate. The implicit assumption holds that what the code does is sufficient specification. It is not: code does not carry governance rights, cannot be compared against composition peers through B2.92, and cannot be used by governance actors without specialized access to implementation details.

Over-broad specification. Architects write content-domain specifications with the belief that comprehensive scope language avoids gaps. The reasoning takes the form: “if the specification is broad enough, nothing falls outside it, so boundary enforcement can never be violated.” This condition produces Form 2—vague content-domain. The result is specifications that are large in scope but empty in operational content. Over-broad specification often emerges during entity birth when architects are uncertain about precise domain boundaries and write expansive language to avoid premature constraint. The expansive language then persists through the entity’s lifecycle, accumulating operational scope that the specification cannot govern.

Enforcement overhead deferred. Architects complete content-domain specification at a useful level of specificity but defer authoring the boundary enforcement rules per B2.91 that activate the specification. The reasoning takes the form: “the specification captures the intent;

we will add enforcement machinery when the system is more stable.” This condition produces Form 3—unenforced content-domain. The deferral converts a governance-intended specification into a documentation artifact. Enforcement that was treated as a deferred task tends to remain deferred: once the entity is operational and accepting inputs, adding enforcement rules requires a directed selection event that must now be justified against an already-running system.

4. Operational Consequences

All three forms of the anti-pattern produce four operational consequences. The consequences vary in intensity across forms—Form 1 produces each consequence in its most severe version because specification is entirely absent—but all four are present in all three forms.

Scope drift. Without authored and enforced content-domain boundaries, entity scope expands organically as new use cases arrive. Developers route new inputs to entities that seem to fit, based on their understanding of the entity’s purpose. Over time, the entity’s actual operational territory diverges from any implicit or vague specification it carries. Because boundaries are not authored in a form that generates enforcement events, no governance record marks where scope was extended, when, by whom, or why. The drift is invisible to governance, untraceable through path retraceability per A1.07, and irreversible in the absence of records that would support rollback.

Composition incompatibility invisible. Composition compatibility per B2.92 requires that both entities in a proposed composition carry authored content-domain specifications that can be evaluated against each other for overlap, gap, or conflict at the domain level. When one or both entities carries an absent, vague, or unenforced content-domain, this assessment cannot be performed. Domain conflicts per B2.91 are never registered in substrate because the authored specifications needed to detect them are absent, too imprecise to derive detection from, or not operationally active. Compositions proceed without governance record of domain-level compatibility, and the consequences of domain incompatibility surface as operational failures rather than as substrate-recorded conflicts that governance could inspect and resolve.

Governance blind to operational territory. The A1.08 substrate-as-source-of-truth commitment requires that the authoritative state of entities—including their operational scope—reside in substrate. When content-domain is implicit, vague, or unenforced, operational territory is not in substrate in the form that makes the A1.08 commitment meaningful. Governance actors exercising inspection rights per A1.01 cannot inspect the entity’s actual scope because the entity’s actual scope is not in substrate. Modification rights cannot be exercised over content-domain because there is no authored content-domain to modify. Override rights cannot be applied to boundary decisions because no boundary decisions are recorded as substrate events. The governance rights that the architecture preserves per A1.01

are formally available but operationally empty with respect to content-domain.

Action-feedback ineffective for domain analysis. Action-feedback per B1.12 examines the entity’s action layer—the record of actual inputs processed and outputs produced—to identify behavioral patterns that could inform DNA evolution through directed selection per B1.14. This mechanism requires that the proposing substrate can distinguish inputs that fall within the entity’s authored content-domain from inputs that fall outside it. Without an authored content-domain specification, or with one too vague to produce this distinction, action-feedback analysis cannot classify the entity’s behavioral history against its domain. Patterns that reflect skilled in-domain performance cannot be distinguished from patterns that reflect accidental out-of-domain processing. DNA evolution proposals produced under this condition may incorporate out-of-domain behavioral patterns into the entity’s instinct layer—evolving the entity’s capabilities in directions that governance never authorized and that no authored content-domain specification ever sanctioned.

5. Detection

Four checks identify the Implicit Content-Domain anti-pattern. They are ordered from most general to most specific.

B2.90 content-domain specification requirements check. For each entity in scope, verify that all required specification elements per B2.90 are present at the appropriate level: input domain schemas for cells, purpose statements for aspects, integration scope for Selves. Absence of any required element is a direct indicator of Form 1. Presence of a specification element that fails the B2.90 specificity check—that cannot answer “is input X in or out of this entity’s domain?”—is a direct indicator of Form 2.

B2.93 content-domain verification. Run the B2.93 verification sequence: specification completeness check, boundary coherence check, and composition compatibility verification. Failure at specification completeness confirms Form 1. Passage at specification completeness followed by failure at boundary coherence—the specification exists but internal boundaries are inconsistent—is a diagnostic indicator of Form 2. Passage at boundary coherence followed by failure at composition compatibility—the specification is internally coherent but cannot be evaluated against composition peers—indicates either Form 2 or Form 3.

B2.91 boundary enforcement rules check. For each entity whose content-domain specification passes the specificity check, verify that boundary enforcement rules per B2.91 are present in entity DNA. Absence of boundary enforcement rules where a specific content-domain specification exists is a direct indicator of Form 3. Absence of boundary enforcement rules where a vague specification exists confirms Form 2—the specification’s imprecision has rendered enforcement rule authoring impossible.

B2.92 composition compatibility assessment. For any proposed or existing composition, verify that content-domain specifications for all entities in the composition can be evaluated against each other for domain-level compatibility. Inability to perform this assessment—because one or more entities carries an absent, vague, or unenforced content-domain—identifies the anti-pattern in the composition context and indicates which entities require remediation before the composition can be governed.

6. Remediation

Remediation tracks the form. The target state in all three cases is an authored content-domain specification specific enough to derive boundary enforcement rules from, with boundary enforcement rules present in entity DNA and confirmed through B2.93 verification.

For Form 1 — Absent content-domain. Author content-domain specifications through directed selection per B1.14, satisfying the B2.90 level-specific requirements for the entity type: input domain schemas for cells authored into DNA per B2.25, purpose statements for aspects, integration scope for Selves. The directed selection event establishing the specification should be recorded as a DNA evolution event under the authority architecture per B1.15, preserving the governance record of when content-domain was established and under whose authority. After the specification is authored, proceed to Form 3 remediation to establish boundary enforcement rules.

For Form 2 — Vague content-domain. Sharpen the existing specification through directed selection until it reaches the specificity required by B2.90: the specification must be able to answer, for each candidate input class, whether that class is in or out of the entity's domain. The sharpening process should surface the implicit knowledge that architects hold about the entity's actual scope and author it into substrate in operational form. Where architects disagree about scope during this process, that disagreement is a content-domain conflict that should be recorded as substrate-level conflict per A1.03 rather than resolved implicitly. After the specification reaches B2.90 specificity, proceed to Form 3 remediation.

For Form 3 — Unenforced content-domain. Author boundary enforcement rules in entity DNA per B2.25 from the existing specific content-domain specification. The enforcement rules translate the authored domain specification into execution-time checks: conditions under which an input is classified as out-of-domain, handling rules for out-of-domain inputs per B2.90, and violation recording requirements per A2.40 for inputs that cross authored boundaries. The enforcement rules are authored through directed selection per B1.14 under the same authority architecture that governs DNA evolution.

For all forms. After remediation, run B2.93 content-domain verification to confirm that specification completeness, boundary coherence, and composition compatibility verification all

pass. Re-run B2.92 composition compatibility assessment for any compositions that involve the remediated entity. Where prior action-feedback analysis was performed without a functional content-domain specification, treat the analysis results as ungoverned with respect to domain classification—do not incorporate behavioral patterns from that analysis into DNA evolution proposals without first establishing which patterns reflect in-domain versus out-of-domain processing.

Summary

	Form 1	Form 2	Form 3
Name	Absent Content-Domain	Vague Content-Domain	Unenforced Content-Domain
Specification present?	No	Yes (imprecise)	Yes (specific)
Boundary rules present?	No	No (cannot derive)	No (not authored)
Primary B2.90 failure	Completeness	Specificity	Enforcement
Primary emergence	Scope as impl. detail	Over-broad specification	Enforcement overhead
Detection check	Specification absent	Specificity check fails	Enforcement rules absent
Remediation path	Author spec → Form 3	Sharpen spec → Form 3	Author enforcement rules

The Implicit Content-Domain anti-pattern is the failure to make operational territory a governed substrate artifact. All three forms produce the same structural outcome: governance cannot see, cannot modify, and cannot enforce the entity’s actual scope—because that scope is not in substrate in operational form. The remediation is consistently architectural: bring content-domain into substrate through directed selection, derive boundary enforcement rules from it, and confirm through B2.93 verification that the specification is complete, coherent, and compatible with composition peers.

Derivation note B3.19 in the CKS Series B defensive publication series. This note formalizes the Implicit Content-Domain anti-pattern as public prior art derived from Paper 2 of the CKS theory series.